

L4.1 Adversarial Attacks Quiz ANS

1. A traffic-sign classifier is trained on sunny daytime photos and deployed in heavy snow, where accuracy drops. How does the lecture classify this?

- A) A white-box adversarial attack, because the snow changes pixel values inside an ℓ_∞ norm ball
 - B) A targeted attack, because the sign is always misread as one specific wrong class by the model
 - C) A naturally occurring distribution shift: no attacker is needed, yet the i.i.d. assumption fails
 - D) A black-box attack, because the weather conditions are unknown to the designer of the model
- ANS: C) A naturally occurring distribution shift: no attacker is needed, yet the i.i.d. assumption fails

2. For the perturbation $\delta = (3, -4, 12)$, which triple gives $(\ell_1, \ell_2, \ell_\infty)$?

- A) (11, 13, 12)
- B) (19, 13, 12)
- C) (19, 169, 12)
- D) (19, 13, 19)

ANS: B) (19, 13, 12)

3. Perturbation A changes two pixels by 7 each, $\delta = (7, 7)$; perturbation B changes one pixel by 10, $\delta = (10, 0)$. Which comparison is correct?

- A) The ℓ_∞ sizes are equal at 10, while ℓ_2 ranks perturbation A as clearly the larger of the two
- B) ℓ_2 sizes are nearly equal (≈ 9.9 vs 10), but ℓ_∞ is 7 vs 10, so B looks far larger under ℓ_∞
- C) Perturbation A is larger under both norms, because it changes more pixels than B does here
- D) The ℓ_2 sizes are 14 vs 10, so A is larger, and ℓ_∞ cannot be computed for a single pixel

ANS: B) ℓ_2 sizes are nearly equal (≈ 9.9 vs 10), but ℓ_∞ is 7 vs 10, so B looks far larger under ℓ_∞

4. What is the ℓ_∞ ball $\{\delta : \|\delta\|_\infty \leq \epsilon\}$ for a 2-pixel image, and why does this matter for FGSM?

- A) A circle of radius ϵ ; FGSM moves along the radius that points toward the gradient direction
- B) A diamond with corners at $\pm\epsilon$ on each axis; FGSM changes only the pixel with the largest gradient
- C) A line segment of length ϵ ; FGSM can change only one pixel by one step of that size
- D) A square with side 2ϵ ; its corners move every pixel by exactly $\pm\epsilon$, which is where FGSM ends up

ANS: D) A square with side 2ϵ ; its corners move every pixel by exactly $\pm\epsilon$, which is where FGSM ends up

5. Model training and adversarial example generation both use gradients. How do their updates differ?

- A) Training ascends the loss in θ ; the attack descends the loss in δ so the output matches the label
- B) Training descends the loss in θ with x fixed; the attack ascends the loss in δ with θ fixed
- C) Both descend the loss, but the attack uses a much larger learning rate and updates θ as well
- D) Training updates the input x to fit the labels; the attack updates θ so that the loss grows

ANS: B) Training descends the loss in θ with x fixed; the attack ascends the loss in δ with θ fixed

6. In PGD, with respect to which variable is the gradient of the loss taken, and why?

- A) With respect to δ : the image x is a constant, and the perturbation is the optimization variable
- B) With respect to θ , because the attacker retrains the network a little at every step of the loop
- C) With respect to the label y , because the attacker moves the label toward the chosen target class
- D) With respect to nothing, because only the sign of the loss value is used in each iteration here

ANS: A) With respect to δ : the image x is a constant, and the perturbation is the optimization variable

7. FGSM with $\epsilon = 0.1$. The loss gradient at $\delta = 0$ is $(-3.0, 0.5, 0.0)$. What is the FGSM perturbation δ ?

- A) $(-0.3, +0.05, 0)$
- B) $(-0.1, +0.1, +0.1)$
- C) $(-0.1, +0.1, 0)$
- D) $(+0.1, -0.1, 0)$

ANS: C) $(-0.1, +0.1, 0)$

8. Why does the exact magnitude of the gradient not matter for FGSM?

- A) A very large step is clipped back to $\pm\epsilon$ in each coordinate, so only the gradient sign survives
- B) The loss is linear in δ , so the gradient magnitude cancels out inside the softmax operator at once

- C) The gradient is always normalized to unit ℓ_2 length before any update is applied to the vector δ
- D) The magnitude matters only for targeted attacks and is irrelevant for an untargeted attack here

ANS: A) A very large step is clipped back to $\pm\epsilon$ in each coordinate, so only the gradient sign survives

9. PGD uses $\epsilon = 8/255$ and a step size $\alpha = 2/255$. How many iterations does the rule of thumb suggest, and why not just one?

- A) A few times $\epsilon/\alpha = 4$, e.g. 10 to 20; one step of size α cannot reach the ball's boundary
- B) Exactly one, because the single-step FGSM direction is already the strongest attack possible
- C) About 255, one iteration for each of the possible pixel intensity levels of the image data
- D) Exactly two, because α is $2/255$ and each iteration must change two pixels at a time by α

ANS: A) A few times $\epsilon/\alpha = 4$, e.g. 10 to 20; one step of size α cannot reach the ball's boundary

10. Compared with FGSM at the same ϵ , what does PGD with small steps typically achieve?

- A) The same loss with a smaller perturbation, using one gradient evaluation like FGSM does here
- B) A perturbation that leaves the ϵ -ball, since projection is applied only once at the very end
- C) A higher loss inside the same ϵ -ball, paid for with several gradient evaluations, not one
- D) Exactly the same δ as FGSM, because the projection at each step erases all of the differences

ANS: C) A higher loss inside the same ϵ -ball, paid for with several gradient evaluations, not one

11. The true class is the first one. The logits change from $(2, 0, 0)$ to $(0, 2, 0)$. How does the cross-entropy loss, $\log \sum \exp(h_j) - h_y$, change?

- A) It falls from about 2.24 to about 0.24, because the largest logit now sits at the wrong class
- B) It rises from 0 to 2, because the softmax outputs sum to 2 after the logits are exchanged
- C) It rises from about 0.24 to about 2.24: the log-sum-exp is unchanged, but $-h_y$ grows by 2
- D) It stays at about 0.24, because the sum of the logits did not change after the exchange

ANS: C) It rises from about 0.24 to about 2.24: the log-sum-exp is unchanged, but $-h_y$ grows by 2

12. Which objective, written as a minimization over δ , describes a targeted attack toward class t ?

- A) Minimize $h_y(x+\delta)$ alone: only the true-class logit matters, and the target comes out by luck
- B) Maximize $h_y(x+\delta) - h_t(x+\delta)$: raise the true-class logit relative to the target-class logit
- C) Minimize $\|\delta\|$ with no reference to any logit: the smallest perturbation is always sufficient
- D) Minimize $h_y(x+\delta) - h_t(x+\delta)$: lower the true-class logit while raising the target-class logit

ANS: D) Minimize $h_y(x+\delta) - h_t(x+\delta)$: lower the true-class logit while raising the target-class logit

13. In the physical stop-sign attack, why is the loss averaged over k transformed copies of the sign image?

- A) To estimate the gradient of a black-box model from the noisy queries sent to an online service
- B) So one δ keeps working across the viewing angles, distances and lighting that a camera will see
- C) To shrink the ℓ_p norm of δ , because averaging cancels out most of the individual pixel changes
- D) To enforce the ℓ_∞ bound without a projection step, because averaging clips each pixel value

ANS: B) So one δ keeps working across the viewing angles, distances and lighting that a camera will see

14. Why is an ℓ_1 term on $M_x \cdot \delta$ used when the perturbation is localized to parts of the sign?

- A) ℓ_1 equals ℓ_∞ for a binary mask, so the ϵ bound is enforced exactly in every masked pixel here
- B) ℓ_1 promotes sparsity, so the change concentrates in a few of the most vulnerable regions
- C) ℓ_1 penalizes large single pixels most, so the change is spread evenly over the whole sign area
- D) ℓ_1 makes the objective differentiable, whereas ℓ_2 would make the loss non-differentiable there

ANS: B) ℓ_1 promotes sparsity, so the change concentrates in a few of the most vulnerable regions

15. What does a robust model's loss surface look like, according to the lecture, and why does it help the defender?

- A) Flatter because capacity is reduced; fewer parameters mean fewer directions for an attacker to use
- B) Non-differentiable, so no gradient exists and gradient-based attackers cannot take any step at all
- C) Identical to the standard model's, with robustness coming only from a larger ϵ used at test time
- D) Smoother: small gradients near the correct label and large distances to other labels

ANS: D) Smoother: small gradients near the correct label and large distances to other labels

16. An attacker owns the target model's training set but not its weights. Which strategy does the lecture describe?

- A) Query the target with random inputs until its API returns the gradients of its own loss function
- B) Recover the target's weights by inverting its softmax outputs on the known training data set
- C) Train a proxy white-box model, attack it, and rely on the attacks transferring to the target
- D) Perturb the training set itself, so the deployed target loses accuracy without any transfer

ANS: C) Train a proxy white-box model, attack it, and rely on the attacks transferring to the target

17. An attacker has neither the training set nor the weights of a cloud model. What does the lecture suggest, and what is the cost?

- A) Collect input–output pairs by querying the service to train a proxy; this may be expensive
- B) Guess the architecture from the vendor name and train it; the cost is a lower final accuracy
- C) Use the gradients returned by the service API; the cost is zero because gradients are public
- D) Give up, because a usable proxy cannot be built without the original training dataset itself

ANS: A) Collect input–output pairs by querying the service to train a proxy; this may be expensive

18. How does a Phantom attack on an ADAS differ from the ϵ -ball perturbations earlier in the lecture?

- A) It shifts LiDAR returns by a bounded ϵ so that the range estimates move inside an ℓ_∞ norm ball
- B) It rewrites the ADAS model weights through a firmware update delivered by a roadside radio unit
- C) It adds ℓ_2 -bounded noise to every pixel of every camera frame so humans cannot see it at all
- D) A depthless picture shown for a few frames, with no ℓ_p bound, placed where viewers do not look

ANS: D) A depthless picture shown for a few frames, with no ℓ_p bound, placed where viewers do not look

19. In the objective $\min_\theta E[\max_\delta \text{Loss}(x+\delta, y; \theta)]$, which gradient updates θ , and what does Danskin's theorem justify?

- A) Inner: maximize over δ to craft the worst-case input; outer: minimize over θ to train on it
- B) Inner: minimize over θ to fit the labels; outer: maximize over δ to craft the attack input here
- C) Inner: the softmax over the logits; outer: the cross-entropy summed over the classes for input x
- D) Both parts are minimizations, one over θ and one over δ , and both are solved at the same time

ANS: A) Inner: maximize over δ to craft the worst-case input; outer: minimize over θ to train on it

20. In the adversarial-robust objective $\min_\theta E[\max_\delta \text{Loss}(x+\delta, y; \theta)]$, which part is the inner and which the outer problem?

- A) The gradient of the loss at the clean input x , because the inner max is constant with respect to θ here
- B) The average gradient over all δ in the ball, weighted by the loss that each of those δ produces in turn
- C) The gradient with respect to δ at δ^* , which is zero at the maximizer, so the parameters θ stay unchanged
- D) The gradient of the loss at $x + \delta^*$, with δ^* the exact inner maximizer; the max need not be differentiated

ANS: D) The gradient of the loss at $x + \delta^*$, with δ^* the exact inner maximizer; the max need not be differentiated